MILITARY SPECIFICATION

Wire Rope Assembly, Mooring, Mark 1 Mod 1

This specification is approved for use by the Naval Sea Systems Command Department of the Navy, and is available for use by Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This Specification establishes the requirements for the procurement the Wire Rope Assembly Mooring Mark 1 Mod 1, LD 549741• It is of one type and class.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following document, of the issue in effect on the date of invit at ion for bids or request for proposal, fom a part of this specification,

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (Code Ident 10001)

LD 549741	Wire Rope Assembly, Mowing, Mark 1 Mod 1.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DL 2424109	Container Unit-Wire Rope Mark 1 Type Assembly.
2428610	Wire Rope Assembly, Mooring, Mark 1, Mod 1

All drawing numbers referenced herein are listed on LD 549741 unless othemise designated.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND Code Ident 10001)

OD 10511	Nonmagnet ic Testing with the Test Set Mark 115 Mod 4 & 5

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publication, The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

HANDBOOK H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents Government Printing Office, Washington, D. C. 20402).

3• REQUIREMENTS

3.1 Description. The Wire Rope Assembly, Mooring, Mark 1 Mod 1, is a nonmagnetic, phosphor bronze, two-component cable with the necessary fittings to attach the cable for end use.

3.2 First article sample Unless otherwise specified in the contract or order, (see 6.2) a first article sample of five wire rope assemblies shall be manufactured and prepared for delivery using the methods and procedures proposed for the production lot. The sample shall be prepared for delivery in accordance with section 5 herein and will be tested for approval as specified in section 4 herein. the sample is for the purpose of determining that, prior to starting production, the contractor's production metbods are capable of producing items that comply with this specification and the design and my be waived by the contracting officer if satisfctory production has been recently accomplished.

3.3 General requirement

3.3.1 Fabrication, The wire rope assembly shall be manufactured in accordance with ID 549741 and as specified herein.

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3.3.2 Interchangeability Component parts of assembled units (unless othewise specified on the applicable drawings fabricated in accordance with applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection for fit or performance.

3.3,3 Threads Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28.

3.4 Magnetic characteristic. The static magnetic effeet of the wire rope assembly shall not be great enough to cause a total change in flux density of the background magnetic field of more than 0.10 milligauss (peak to trough)

3.5 Cleanup. Prior to and after final assembly, all parts components and the assembly shall be thoroughly cleaned of metal chips and other foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perfrom , any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.2 Classification of inspection. Inspection of wire rope assembly shall be classified as follows:

a. first article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)
c. periodic production inspection (see 4.6)

4.3 Test equipment . The following item of test equipment is required to perform the tests required in this specification:

Test Set Mark 115 type with Adapter Mark 36 Mod O and Magnetic Effects Test Stand Mark 9 Mod O (Government furnished).

4.4 First article inspection.

4.4.1 First article testing. The first article sample of five (5) wire rope assemblies (see 3.2) shall be subjected to the examinations and tests detailed in 4.7 (see Table I) at an activity designated by the procuring activity. Further production of the wire rope assembly by the supplier, prior to approval of the sample, shall be at the supplier's risk.

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4,4.2 First article sample approval. First article sample approval will be granted when the Government testing act ivit y, through testing and subsequent analysis of the test results, is assured of contractor conformance to the contract technical requirements. Conditional acceptance may be granted at the option of the Government based on analysis of detected defects. A first article sample may be disapproved when either nonconforming character-istics or unauthorized materials are found.

4.5 Quality conformance inspection (see notes, Table I).

4.5.1 100% inspection. Each unit shall be subjected to the examinations and test (designated as 100) in 4.7,1 and 4.7.2. Any unit containing one or more defects shall be cause for rejecting the unit.

4.5.2 Sampling. The tests of 4.7.3 (designated as sampling shall be conducted on a random sample of assemblies selected from each production of 1.0 percent. Failure of the sample to meet this requirement shall cause the lot to be rejected. Lot composition shall be in accordance with mil-std-105 Section 5.

4.6 Periodic production inspection, Periodic production tests consisting of the tests of 4.7 (as specified in Table I), shall be conducted by a designated Government Testing activity on a random sanple of two packaged wire rope assemblies, unless otherwise specified (see 6.2.h), selected f from each production lot offered for accept ante. When a unit fails to pass any periodic production test, such failure shall be cause for rejection of the periodic sample and the lot it represents.

4.7 Examination and test methods.

4.7.1 Preparation for delivery (100%). The packaging shall be examined to detemine conformance to section 5 herein.

4.7.2 Visual examination (100%). Each wire rope assembly shall be visually examined in accordance with drawing 2428610 to determine compliance with this specificat ion and the applicable drawings, Examination of the assembly in accordance with drawing 2428610 shall not relieve the contractor of the responsibility for meeting the requirements of other drawings listed thereon.

4.7.3 Magnetic effeds testing (Sampling Perform the magnetic effects test on the wire rope assemly in accordance with test Method 1, OD 10511, Perform the test with the assembly packaged as specified in section 5. The test face shall be one of the large, open-side faces of the unit container, The distance from the center of the magnetometer to the test face of the container shall be 5 ± 1/8 inches. The rate at which the container shall be rotated while conducting the test of OD 10511, Method 1, orientation (C), shall be between 5 and 15 seconds per revolution. Verify that the assembly does not produce a magnetic signal exceeding 0.10 milligauss.

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TABLE I. Correlation of requirement and test method paragraphs.

Title of Examination or Test	Requirement Paragraph	Method Paragraph	Inspection Classifications (1) First (2) Article	Quality (3) Conformance	Periodic (2) Production
Packaging	5	4.7.1	X	X	X
Visual	3.3.1, 3.3.2 3.3.3, 3.5 and 3.6	4.7.2	X	X	X
Magnetic Effects Testing	3.4	4.7.3	X	X	X

TABLE I NOTES

(1) The X in each column indicates that the tests are to be performed as appropriate in accordance with notes (2) and (3) below.

(2) Tests for the first article and periodic production samples shall be performed by a designated Government testing activity.

(3) Quality conformance inspection shall be an in-process function performed by the manufacturer during assembly and preparation for delivery of the wire rope assemblies.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A.

5.1.1 Level A. Each wire rope assembly shall be individually packaged in accordance with DL 2424109 unless otherwise specified in the contract or order.

5.2 Marking. In addition to any special marking requied by the contract or order all marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. the wire rope assembly is intended for use in undewater mines to provide a nonmagnetic link between a magnetic mine and the steel moring cable.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Number of first articles required (see 3.2).

c, Test equipment to be loaned by the Government it, and the terns under which it is loaned (see 4.3a).

d. Activity to perform first article testing.

e. Act ivity to perfom periodic production inspection.

f. Lot size for sampling purposes if one month's scheduled production is not acceptable (see 6. 3).

g. special marking when required.

h. Nunber of units in the periodic sample if other than specified (see 4.6).

6,3 Lot, Unless othewise specified in the contract or purchase order, lot size shall mean one ninth's scheduled production. See sect ion 5 of MIL-STD-105 for lot compositions

6.4 Design agency , The design and technical cognizance for the assembly specified herein are located at the Naval Mine Engineering Facility Yorktown, VA 23691.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes .

Custodian:
NAVY - OS

Preparing activity:
NAVY - OS

Project Number
1350-NO34

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-W-21993B(OS) Wire Rope Assembly, Mooring, Mark 1 Mod 1

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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